ANALYSIS REPORT

Track 1 SVM Exact-Faithful Final Attempt Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the final strict paper-faithful `SVR` attempt prepared in:

- `doc/reports/campaign_plans/2026-04-17-11-44-20_track1_svm_exact_faithful_final_attempt_campaign_plan_report.md`

  The campaign executed `4` exact-paper validation runs through the stabilized remote launcher:
- completed runs: `4`
- failed runs: `0`
- execution window: `2026-04-17 12:00:42+02:00` to `2026-04-17 17:36:26+02:00`
- campaign artifact root:

  `output/training_campaigns/track1_svm_exact_faithful_final_attempt_campaign_2026_04_17_11_44_20/` The campaign goal was intentionally narrow:
- rerun the residual `SVM` yellow cells under the exact same recovered

  paper-faithful `SVR` path;
- avoid any algorithm change or widened hyperparameter search;
- determine whether one last exact-faithful attempt could close `40`, `240`,

  or `162`.

Objective And Outcome

The campaign had four concrete questions

1. does the exact-faithful amplitude pair rerun improve the combined `40/240` residual surface?
2. does an isolated exact-faithful rerun close amplitude harmonic `40`?
3. does an isolated exact-faithful rerun close amplitude harmonic `240`?
4. does an isolated exact-faithful rerun close phase harmonic `162`?

Outcome:

- the remote launcher completed the full campaign end to end and synchronized all validation artifacts back to the repository;
- all `4` scoped validation runs completed successfully;
- every rerun reproduced the same known exact-paper `SVR` values already seen in the previous reference-grid campaign;
- none of the residual paper targets reached or beat the paper numeric target;
- all scoped `SVR` ONNX exports still failed under the current stack, but the runs completed because `export_failure_mode` remained `continue`. This means the campaign succeeded operationally, but scientifically it confirmed the exact-faithful `SVR` plateau rather than closing any new paper cell.

Ranking Policy

This campaign is not a full family-matrix comparison and it is not a true closure campaign, because every run stayed above its scoped paper target.

The serialized campaign policy is therefore

- primary metric: `mean_normalized_gap_ratio_asc`
- first tie breaker: `max_normalized_gap_ratio_asc`
- second tie breaker: `open_paper_cell_count_asc`
- third tie breaker: `failed_target_count_asc`
- fourth tie breaker: `run_name`

Where:

- `normalized_gap_ratio = (repository_metric - paper_metric) / paper_metric`
- lower is better
- the phase run uses both paper cells available on harmonic `162`: `MAE` and `RMSE`
- the amplitude runs use the paper `RMSE` cells available for harmonics `40` and `240` Interpretation:
- this policy exists only to serialize the required campaign-best artifacts;
- the scientific reading remains that all scoped exact-faithful `SVR` targets are still open.

Campaign Ranking

Ranked Completed Runs

Rank	Run	Scope	Paper Cells	Mean Gap Ratio	Max Gap Ratio	Failed Exports
1	`track1_svr_exact_faithful_phase_162_repeat`	`phases_only`	2	1.307	1.711	1
2	`track1_svr_exact_faithful_amplitude_40_repeat`	`amplitudes_only`	1	1.676	1.676	1
3	`track1_svr_exact_faithful_amplitude_pair_repeat`	`amplitudes_only`	2	7.745	13.814	2
4	`track1_svr_exact_faithful_amplitude_240_repeat`	`amplitudes_only`	1	13.814	13.814	1

Export Surface

Completed Runs	Exported ONNX Files	Failed Exports	Surrogate Exports
4	0	5	0

The launcher path is therefore stable, but the exact-paper `SVR` ONNX export path is still not operational under the current environment.

Campaign Best Run

The explicit bookkeeping winner is

- `track1_svr_exact_faithful_phase_162_repeat`

  It was selected because it achieved:
- the lowest scoped mean normalized paper-gap ratio: `1.307`
- the lowest scoped max normalized paper-gap ratio among the four runs
- the cleanest remaining paper-gap approach under the strict exact-faithful

  rerun constraint Important interpretation:
- this run is the campaign bookkeeping winner;
- it is **not** a closure run;
- both scoped paper cells for phase `162` remain above the paper target.

Scientific Outcome

Scoped Target Metrics

Run	Best Params	Scoped Result
`track1_svr_exact_faithful_amplitude_pair_repeat`	`C=0.1` , `epsilon=1e-05` , `kernel=rbf`	`A40` MAE `8.20e-05` ; RMSE `9.63e-05` and `A240` MAE `3.02e-04` ; RMSE `6.22e-04`
`track1_svr_exact_faithful_amplitude_40_repeat`	`C=0.001` , `epsilon=0.0001` , `kernel=rbf`	`A40` MAE `8.20e-05` ; RMSE `9.63e-05`
`track1_svr_exact_faithful_amplitude_240_repeat`	`C=0.1` , `epsilon=1e-05` , `kernel=rbf`	`A240` MAE `3.02e-04` ; RMSE `6.22e-04`
`track1_svr_exact_faithful_phase_162_repeat`	`C=1.0` , `epsilon=1e-06` , `kernel=rbf`	`phi162` MAE `0.542246` ; RMSE `1.217833`

Gap Versus Paper

Harmonic / Metric	Paper Target	Repository Result	Gap	Status
`A40` `RMSE`	`3.60e-05`	`9.63e-05`	`6.03e-05`	above paper target
`A240` `RMSE`	`4.20e-05`	`6.22e-04`	`5.80e-04`	above paper target
`phi162` `MAE`	`0.200`	`0.542246`	`0.342246`	above paper target
`phi162` `RMSE`	`0.640`	`1.217833`	`0.577833`	above paper target

What The Campaign Actually Established

- the exact-faithful rerun of `40` reproduces the same known `SVR` result;
- the exact-faithful rerun of `240` reproduces the same known `SVR` result;
- the exact-faithful rerun of `162` reproduces the same known `SVR` result;
- the pair rerun does not unlock any coupled improvement over the isolated results;
- the strict exact-faithful `SVR` path therefore appears plateaued on the last residual `SVM` yellow cells.

Operational Outcome

The campaign still delivered a meaningful closeout result.

It proves that the repository now has:

- a stabilized remote exact-paper launcher;
- a fully repeated exact-faithful `SVR` rerun package on the residual cells;
- a second independent confirmation that the remaining `SVR` gaps are not being caused by missing paper-grid execution. That makes the negative result scientifically stronger than the first reference-grid pass alone.

Canonical Benchmark Impact

This campaign does **not** promote any new canonical paper-cell closure into `Track 1`.

So the canonical benchmark reading remains unchanged:

- no new Tables `2-5` cells close because of this campaign;
- no new Table `6` harmonic becomes fully matched;
- the latest result is a plateau confirmation, not a closure upgrade.

Main Conclusions

The campaign supports four conclusions.

1. Exact-Faithful SVR Is Now Repeatedly Reproduced

The repository has now repeated the paper-faithful `SVR` route on the residual targets through a second strict campaign. The remaining gap is therefore no longer plausibly explained by "we still have not really run the paper `SVR` path."

2. The Residual SVM Yellow Cells Remain Open Under Faithful Reproduction

Residual harmonics `40`, `240`, and `162` all remain above the paper target even after the final strict rerun package.

3. The Final Attempt Confirms a Practical SVR Plateau

Because the reruns reproduce the same metrics already seen in the earlier reference-grid campaign, the repository should treat the residual `SVM` gap as plateaued under the recovered `SVR` method rather than as an unresolved tuning-surface omission.

4. ONNX Export Remains Open But Secondary To The Plateau Result

Training and validation still run correctly, but all scoped `SVR` ONNX exports failed again. That remains a deployment-facing issue, but it is now secondary to the stronger scientific conclusion that exact-faithful `SVR` itself is no longer closing the row.

Produced Campaign Artifacts

The campaign output folder now contains the required serialized winner artifacts:

- `output/training_campaigns/track1_svm_exact_faithful_final_attempt_campaign_2026_04_17_11_44_20/campaign_leaderboard.yaml`
- `output/training_campaigns/track1_svm_exact_faithful_final_attempt_campaign_2026_04_17_11_44_20/campaign_best_run.yaml`
- `output/training_campaigns/track1_svm_exact_faithful_final_attempt_campaign_2026_04_17_11_44_20/campaign_best_run.md`

The remote wrapper log for the successful campaign execution is:

- `.temp/remote_training_campaigns/2026-04-17-12-00-24_track1_svm_exact_faithful_final_attempt_campaign_2026_04_17_11_44_20/remote_training_campaign.log`

The canonical reports refreshed by this close-out are:

- `doc/reports/analysis/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`